

Component	Description / Spec	Q	Source	Cost	Manufacturing Method
Raw Material: Delrin Plate	Acetal plastic, 1/2" thickness (for Ratchet, Pawl, Cam)	1	McMaster-Carr	\$35.00	Manual Mill / Lathe fabrication
	1/4" thickness		Laser cutting		
Raw Material: Acrylic Sheet	Clear/Black Acrylic, 1/4" thickness (for inner structural plates) 3-5 layers, separate plates for phalanges	1	Engineering Shop	\$15.00	Laser Cutting
Worm & Worm Gear Set	Matched set (e.g., Module 1 or 24 DP, brass or steel)	1	McMaster-Carr	\$45.00	Off-the-shelf (No modification)
Palm Gear	Spur gear to mesh with internal finger drive	1	McMaster-Carr	\$20.00	Off-the-shelf
Miniature Chain & Sprockets	#25 or smaller roller chain + 2 matching sprockets (for finger link)	1	McMaster-Carr	\$30.00	Off-the-shelf, cut chain to length
Precision Axles / Shafts	1/4" or 6mm D-shafting or Keyed Stainless Steel	2	McMaster-Carr	\$12.00	Cut to length on shop bandsaw
Flanged Bearings / Bushings	Shielded ball bearings matching chosen shaft diameter	4	McMaster-Carr	\$5.00	Press-fit into laser-cut acrylic plates
Pawl Spring	Small extension or torsion spring	1-2	McMaster-Carr (Assortment Pack)	\$15.00	Off-the-shelf

Aesthetic Outer Casing	PLA or PETG filament (ergonomic/anatomical hand shell)	1	WEEF / Sedra Student Design Centre	\$10.00	3D Printing (FDM)
Position Motor	Given servo / stepper	1	Provided by Course	\$0.00	None (Mount to chassis)
Finger Drive Motor	Given DC motor	1	Provided by Course	\$0.00	None (Mount to chassis)
Custom Control PCB	Main board for EMG processing	1	JLCPCB / PCBWay (or local rapid proto)	\$20.00	Solder components to custom board

Dijiki, sign up with uwaterloo email

3d printing & assembly this week

Laser cutting > cutmypart.ca

Maybe on campus

Components

- Worm gear x1
 - 3D print
- Main gear in palm (Ratchet)
 - Acrylic laser cut/3D print
- Pawl
 - 3D printed/laser cut
- Axle
 - Order online, then cut
- Small gears in base of finger
 - 3-D print
 - They need a space to hold a chain (or string for rapid prototype)

- Top phalange
 - Lazer cut/3D print
 - It would look like a gear with a stick coming out of it
- Springs x2
 - Online
- Chain
 - Online
- Outside 3D printed

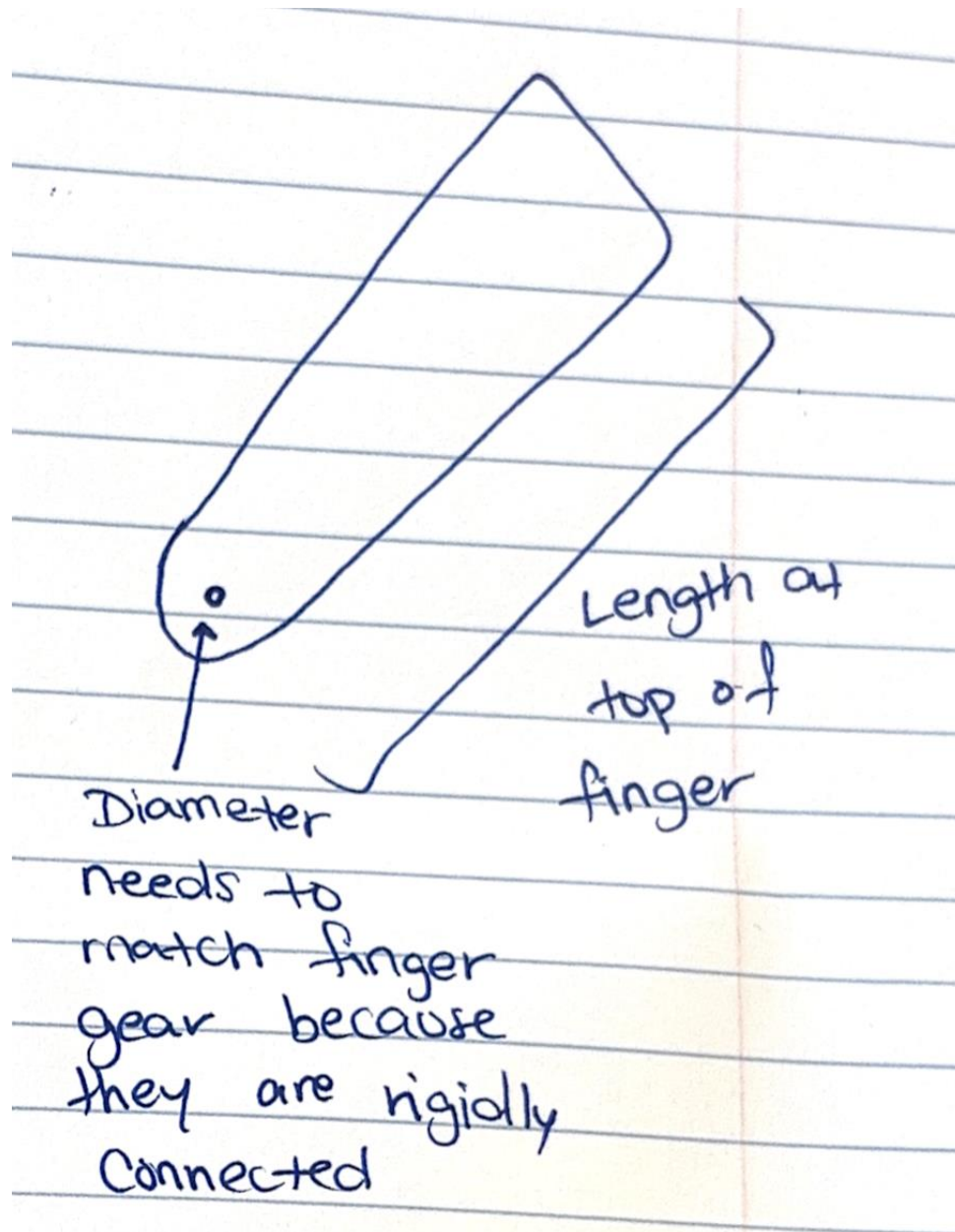
Amazo

- Chain
 - https://www.amazon.ca/PLATT-1-Speed-8-Speed-114-Links-116-Links/dp/B091YYZYD1/ref=sr_1_2_sspa?crid=3GUWKPJ6MFUG1&dib=eyJ2IjoiaMSJ9.ALg4fSI8lqgoGLt2HN42rhUkd8gKPhdiKcRmSP26PWQKzzS0SrRYqGbop-wTBh5MrQS041bLfvrKpxdQdDsgGqx7I3FuIO8237KRkN3iYOiYQAF9715H3j7epdlEypmml_1csLb3_jNVYEJkGUlx-0n33237ZVjhToliPFfC6iSFhGGnXrEaYoaQYYf2-kP3WqwuZVrRAR4t3q_6aLkhZzKNBhGpK61v6ed2-ll8aB6rqNM1Taop29zUSyJNXOUIZrsNKjDL-KrZqw7XhIDG-8WsvGEb6P2Yz-d5Ftq4fA.Zkxq3OnbuVJxu3GqyHoERLmuFnX2OBzIXVjHqthVXxU&dib_ta g=se&keywords=bike%2B11%2Bchain&qid=1784040153&sprefix=bike%2B11%2Bchain%2Caps%2C135&sr=8-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1
 - Make sure to get 11-speed
 - \$23.98
- springs
- Worm gear
- Ratchet
- Pawl
- Small gears in base of finger
- Top phalange

*Things to keep in mind:

- Main axle in hand only rotates about 75 degrees

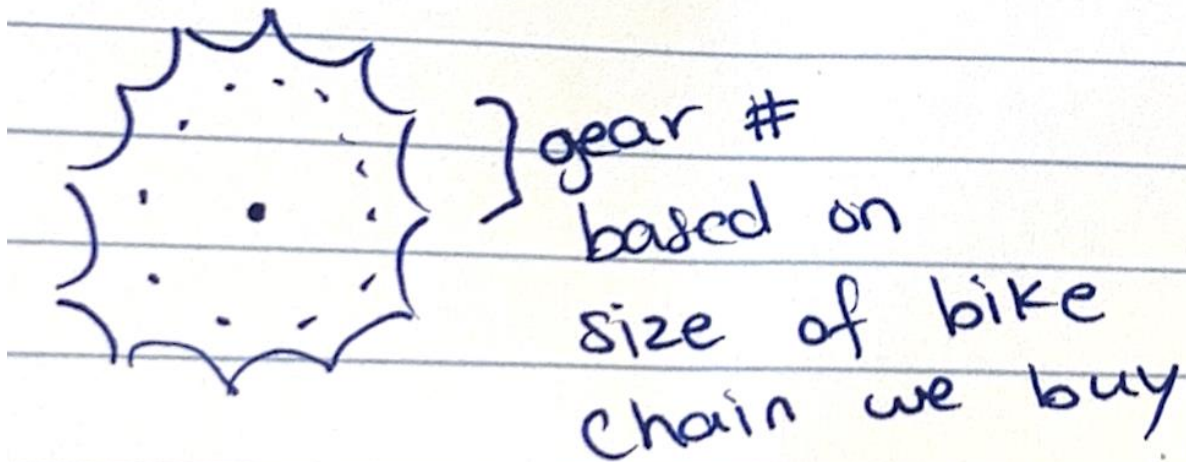
- Worm gear



Hilda

- Finger gears x4 **Note we are buying a 11-speed chain, ensure the dimensions of the gear fit this change

Colme



Adithi

- Ratchet
- pawl

